The Multiplicity of Meaning, chapter 4

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23rd April 2025

1 *Prior's theorem*[1]

[1] **PriorFP**.

A thought experiment: Mr X, a normal English speaker, is in Room 7, but wrongly thinks he is in Room 6 and that a pathological liar is in Room 7. On this basis, Mr X utters

S: Not everything being asserted in Room 7 is true.[2]

[2] Prior has universal quantification instead, which can be used to run a parallel argument: I prefer existential since I think even the most pathological liars often assert truths.

Argument that Mr X is saying more than one thing:[3]

[3] This is 'Argument B' in the current version of the manuscript. The similar 'Argument A' corresponds to Prior's original version.

(1) Someone in Room 7 is asserting that not everything being asserted in Room 7 is true. (Premise)

(2) If everything being asserted in Room 7 is true, then not everything being asserted in Room 7 is true. (From 1)

(3) Not everything being asserted in Room 7 is true. (From 2)

(4) Something being asserted in Room 7 is true. (From 1 and 3)

(5) At least two things are being asserted in Room 7. (From 4 and 5)

One natural approach to formalizing this argument represents quantification over 'things asserted' using *quantification into sentence position*. On this approach, the steps from 1 to 2 and from 1 and 3 to 4 are licensed by the standard 'universal instantiation' and 'existential generalization' rules:

1.	$Q\neg(\forall p.Qp \to p)$	Premise
2.	$(\forall p.Qp \to p) \to \neg(\forall p.Qp \to p)$	1, UI
3.	$\neg(\forall p.Qp \to p)$	2, PC
4.	$\exists p.\, Qp \wedge p$	1, 3, EG
5.	$\exists p \neg\forall p'.\, Qp \wedge Qp' \to p = p')$	3, 4, LL

An alternative approach would treat the status of 'true' as a predicate of *propositions* more seriously, in which case these steps would implicitly appeal to the following

Propositional Truth Schema The proposition that *P* is true iff *P*.

- Some would reject the step from (2) to (3), which fits the following schema:

  CLAVIUS'S LAW If P, then not-P. Therefore, not-P.

- Some will reject premise (1).

  NB: I am not *not* arguing for (1) from some premise like 'Any English speaker who utters S thereby asserts that someone in Room 7 is saying something untrue'. In my view, merely knowing what words someone spoke is almost never sufficient for knowing that they said p, for any p.

  Still, we *do* often have such knowledge. And given the background of the throught experiment, (1) seems a straightforward case of it.[4]

[4] If you are inclined to deny this, consider variant arguments using 'Someone in Room 7 seems to be saying that...' or 'If one thought the person in Room 7 was in Room 6 one would take him to be saying that...', modifying S accordingly.

2 *Same sentence uttered, different propositions asserted*

At step (3), *I* uttered S; but unlike Mr X, *I* didn't say anything false by doing so. What makes the difference?[5]

[5] **PriorFP**: 'How, we all want to cry out, can what a man is thinking and even what a man can be thinking on a given occasion, depend on what number is written on the other side of a door?'

Add Ms Y in Room 8 and and Ms Z in Room 9, both uttering S. Ms Y, like Mr X, falsely believes that there is a pathological liar in Room 7. Ms Z knows what Mr X is uttering in Room 7 and why, and runs through the first three steps of the above proof.

- All three are saying that note everything being asserted in Room 7 is true.
- Ms Z isn't saying anything false.
- Ms Y is saying (roughly) the same things as Mr X, including at least one false.

3 *An utterance that refers to infinitely many properties*

We can push a bit further towards the full strength of Plural Signification using a case modelled after Berry's paradox.[6]

[6] 'The least integer not nameable in fewer than nineteen syllables': reported by RussellMLBTT. (The analogous pluralist solution is that this eighteen-syllable definite description "names" infinitely many integers. So, there *is* no least integer not nameable in fewer than nineteen syllables).

Utterance u **picks out** (natural) number $n := n$ is the least instance of some property u refers to.

Suppose Ms Y, in room 4, only utters:

> [B] Some number is not picked out by any utterance of a verb phrase in room 4.

> There is a verb-phrase-utterance in room 4 that refers to the property of not being picked out by any verb-phrase-utterance in room

4. So for any n, if n is the least number not picked out by any verb-phrase-utterance phrase in room 4, n is picked out by a verb-phrase-utterance in room 4 (from **??** and the definition of 'pick out'). So, there *is* no least number not picked out by any verb-phrase-utterance in room 4 (from **??**). So, every number is picked out by a verb-phrase-utterance in room 4 (from **??** by the least-number principle). So, Ms Y's one verb-phrase-utterance refers to infinitely many properties (from **??** and the setup).

4 *Linguistic meaning*

Here's another instance of Prior's theorem:

> If a sentence expresses the proposition that it expresses a false proposition, then it expresses at least two propositions.

(L): (L) expresses a false proposition.

To avoid the conclusion that (L) expresses multiple propositions, we would have to deny that (L) expresses the proposition that it expresses a false proposition. That looks hard!

To avoid distractions about the semantics of temporary letter-names and the diagonal schema, we can instead consider predicates, and appeal to the following variant of Prior's theorem:

> If a predicate expresses *expressing some property that one lacks*, then it expresses at least two properties (viz. some property that it lacks, as well as at least one property that it has, namely, *expressing some property that one lacks*).[7]

[7] Derivation:

1.$Qx(\lambda y.\exists Z(QyZ \wedge \neg Zy))$	Premise
2.$\forall Z(QxZ \rightarrow Zx) \rightarrow \exists Z(QxZ \wedge \neg Zx)$	1, UI
3.$\exists Z(QxZ \wedge \neg Zx)$	2
4.$\exists Z(QxZ \wedge Zx)$	1, 3, EG
5.$\exists Z \exists Z'(QxZ \wedge QxZ' \wedge Z \neq Z')$	3, 4, LL

But surely 'expresses a property it lacks' expresses *expressing a property one lacks*! So it expresses at least two properties.

5 *Sentential truth*

If we define 'true sentence' as 'sentence that expresses something and expresses only truths', the above claims about (L) carry over to:

(Q): (Q) is not true.

So, (Q) is not true. If we define 'false sentence' as 'sentence that expresses something and expresses only falsehoods', then we also get that (Q) is not false.[8]

With these definitions our view counts as a 'classical gap theory'.[9] The

[8] Given that the negation of a sentence expresses $\neg p$ iff the sentence expresses p, this makes 'false sentence' equivalent to 'sentence whose negation is true'.

[9] **FieldSTFP**.

standard challenge for classical gap theorists is: why are you going around asserting sentences you think aren't true?[10]

Response: asserting *propositions* you think aren't true is bad, but on this definition of 'true', asserting *sentences* you think aren't true ('Tanks were invented during World War I').[11]

[10] See **MaudlinTP** for a response very different from mine.

[11] If we instead defined 'true sentence' as 'sentence expressing at least one truth' the view becomes a 'classical glut theory' where Q is both true and false. Classical glut theories face a dual challenge: what objection would you have if someone were to assert all these sentences that you say are true even though for some reason you hold back from asserting them? Our response is parallel: asserting sentences you think are "true" in this sense can be bad, if you're thereby asserting false propositions.

6 Context-relativity

Here's another instance of the property-variant of Prior's theorem:

(*): If, relative to some $\langle x, t, w\rangle$, a certain predicate expresses *being such that there is a property one lacks which one expresses relative to* $\langle x, t, w\rangle$, then it expresses at least two properties relative to $\langle x, t, w\rangle$.

Now consider the following predicate:

HC: is such that there is a property it lacks and expresses relative to ⟨me, the present time, the actual world⟩.

Here is an argument that it does the thing that, according to (*), implies that it expresses at least two properties relative to every $\langle x, t, w\rangle$:

(i) Relative to any $\langle x, t, w\rangle$, 'me' refers to x.

(ii) Relative to any $\langle x, t, w\rangle$, 'the present time' refers to t.

(iii) Relative to any $\langle x, t, w\rangle$, 'the actual world' refers to w.

(iv) Relative to some $\langle x, t, w\rangle$, 'express relative to' expresses *expressing relative to*.

(v) If, relative to $\langle x, t, w\rangle$, 'me' refers uniquely to x, 'the present time' refers uniquely to t, 'the actual world' refers uniquely to w, and 'express relative to' refers uniquely to R, then relative to $\langle x, t, w\rangle$, HC uniquely expresses *being an z such that there is a property Y such that z lacks Y and $R(z, Y, \langle x, t, w\rangle)$*.

7 Infinitely many things expressed

Definition: Predicate *a picks out* natural number n := a expresses some instantiated by n and nothing else.

(B): is the least number not picked out by any predicate of fewer than 70 characters

Suppose for contradiction that there is a natural number not picked out by a predicate of fewer than 70 characters. Then there is a least such number n. Since n is the only number to instantiate *being the least number not picked out by a predicate of fewer than 70 characters*, and (B) expresses this property, (B) picks out n. But (B) has only 67 characters: contradiction!

Hence every natural number is picked out by a predicate of fewer than 70 characters. This implies that the predicates of fewer than 70 characters between them express infinitely many properties. Since they are finite in number, at least one of them expresses infinitely many properties.

8 *The status of the disquotational meaning schema*

Consider this schema:

DM '*A*' expresses *A*.

Understand this so that the following all count as DM-instances:

(D1) 'Snow is white' means[12] that snow is white

(D2) 'is white' expresses *being white*.

(D3) '(L) expresses a falsehood' means that (L) expresses a falsehood

(D4) 'expresses a property it lacks' expresses *expressing a property one lacks*

They sound good! But some DM-instances, e.g. (D3) and (D4), must express some falsehoods, since they logically imply sentences (like '(L) expresses a falsehood') which express falsehoods.

- My view is that not only 'paradoxical' instances (D3) and (D4), but even boring instances like (D1) and (D2), express falsehoods (as well as truths).

Puzzle: how is it that the negations of DM-instances get to express truths, when they sound so bad?

Observation: Any falsehood expressed by (D1) is of the form M('Snow is white', q), where 'means' expresses M and 'snow is white' means q.[13] So for (D1) to mean something false, 'means' must express some relation M that's narrower than *meaning*: a relation that fails to hold between 'snow is white' and some propositions that it means.

[12] I'll use 'means' for the expressing relation between sentences and propositions.

[13] For simplicity, I'm assuming semantic uniqueness for quote-names.

To develop some intuition for what these narrower meanings might be like, first consider a case of non-literal speech: 'Travolta is a comet', used to convey that Travolta is intermittently famous. Once we have committed to the metaphor, we could go on to…

- say 'Travolta is not a comet' to convey that Travolta is not intermittently famous.
- say 'People think that Travolta is a comet' to convey that people think that Travolta is intermittently famous
- say

  (1) 'Snow is white' doesn't mean that Travolta is a comet

  to convey the truth that 'Snow is white' doesn't mean that Travolta is intermittently famous.
- say

  (2) 'Travolta is a comet' doesn't mean that Travolta is a comet

  to convey the truth that 'Travolta is a comet' doesn't mean that Travolta is intermittently famous.

This helps us see how someone might end up using the negation of a DM-instance to *convey* a truth, though of course in this case, the truth isn't *meant* by the sentence, since the use is non-literal.

But now reflect on the semantic flexibility of the word 'literal' itself. Plausibly, 'literal' expresses both a property that applies to the Hollywood use of 'is a star' and a more demanding property that doesn't. Let's stipulate that in the following argument it's being used in the former way.

If we had gone for the more demanding standard for 'literal', we would also have said:

(3) 'Travolta is a star' doesn't mean that Travolta is a famous actor.

And if we combined the demanding use of 'literal' and 'mean' with the Hollywood use of 'star' we could get the same truths across by saying:

(4) 'Travolta is a star' doesn't mean that Travolta is a star.

Here, as with (7), we are *conveying* a truth (the same one we convey with (8)). But our use of (9) to convey these truths now counts as a *literal* one.[14] So we should accept 'There is a truth among the proposi-

[14] It doesn't have the properties we would *then* refer to with 'literal', but that doesn't matter.

tions meant by (9)′.

This gives us our hoped-for purchase on what the truths meant by the negations of DM-instances are like, and how they end up being expressed despite their somewhat odd-sounding character.